

Regional Near-Surface Air Temperature Regression Analysis

Climate Projections Using CMIP6 Data (SSP1-1.9 & SSP5-8.5)

Climate Change: Impact and Adaptation

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Introduction

Study Area:

A region of 30 x 30 degrees
(20°N ~ 50°N, 90°E ~ 120°E)

Data Source:

Copernicus Climate Data Store (CDS)

EC Earth3 (Europe) for SSP1-1.9;
EC Earth3CC (Europe) for SSP5-8.5.

Data Processing:

Rstudio Linear Regression Model

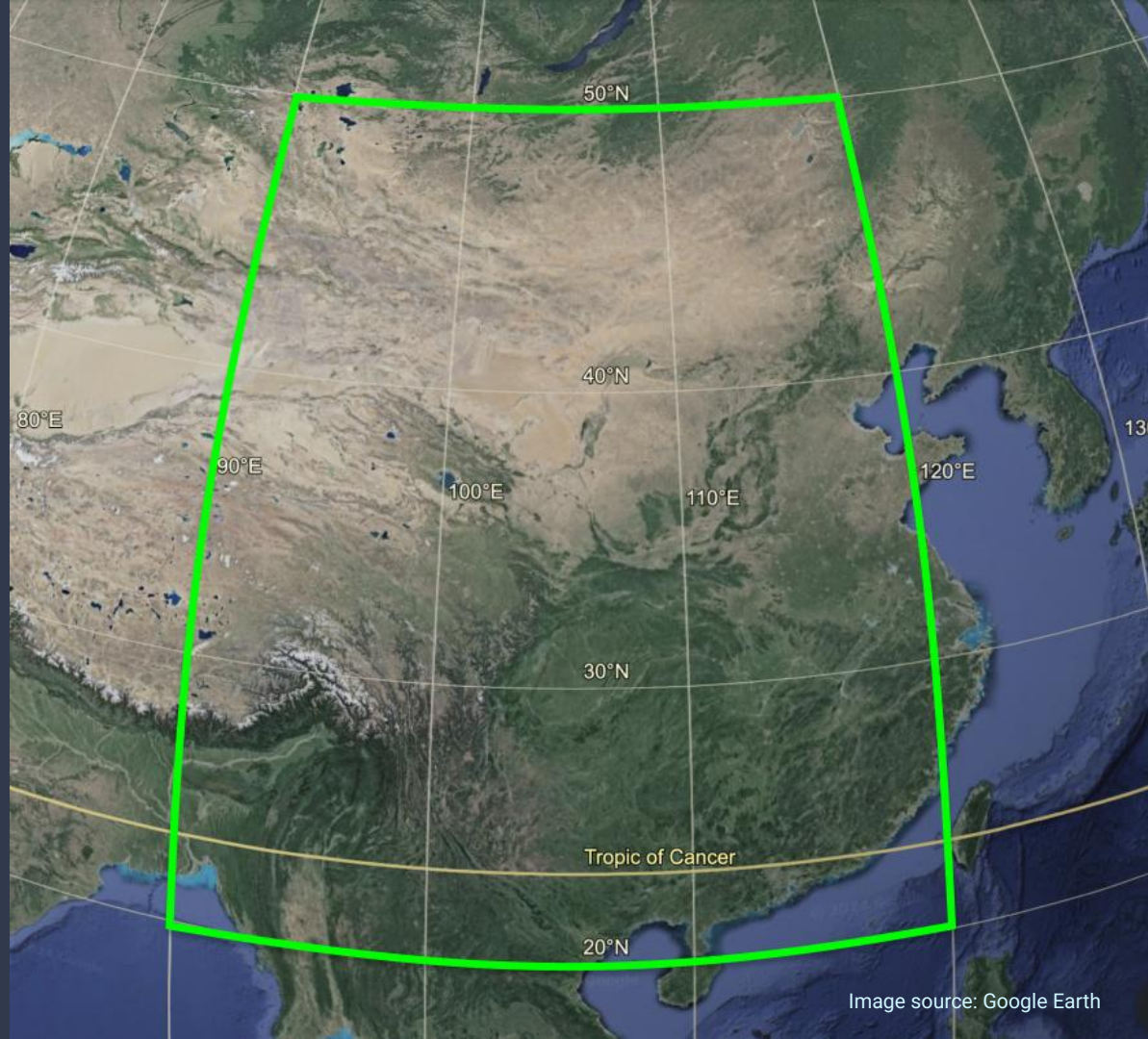
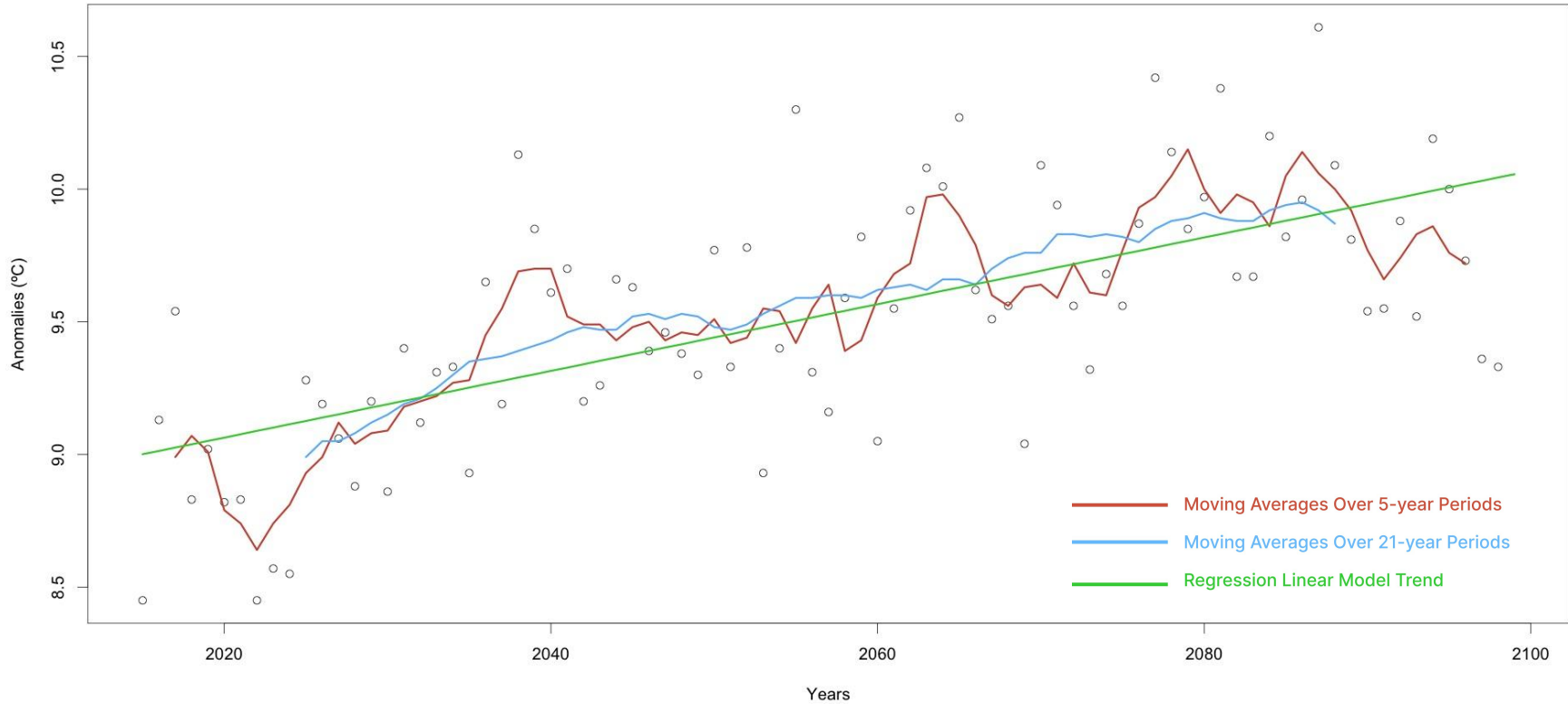


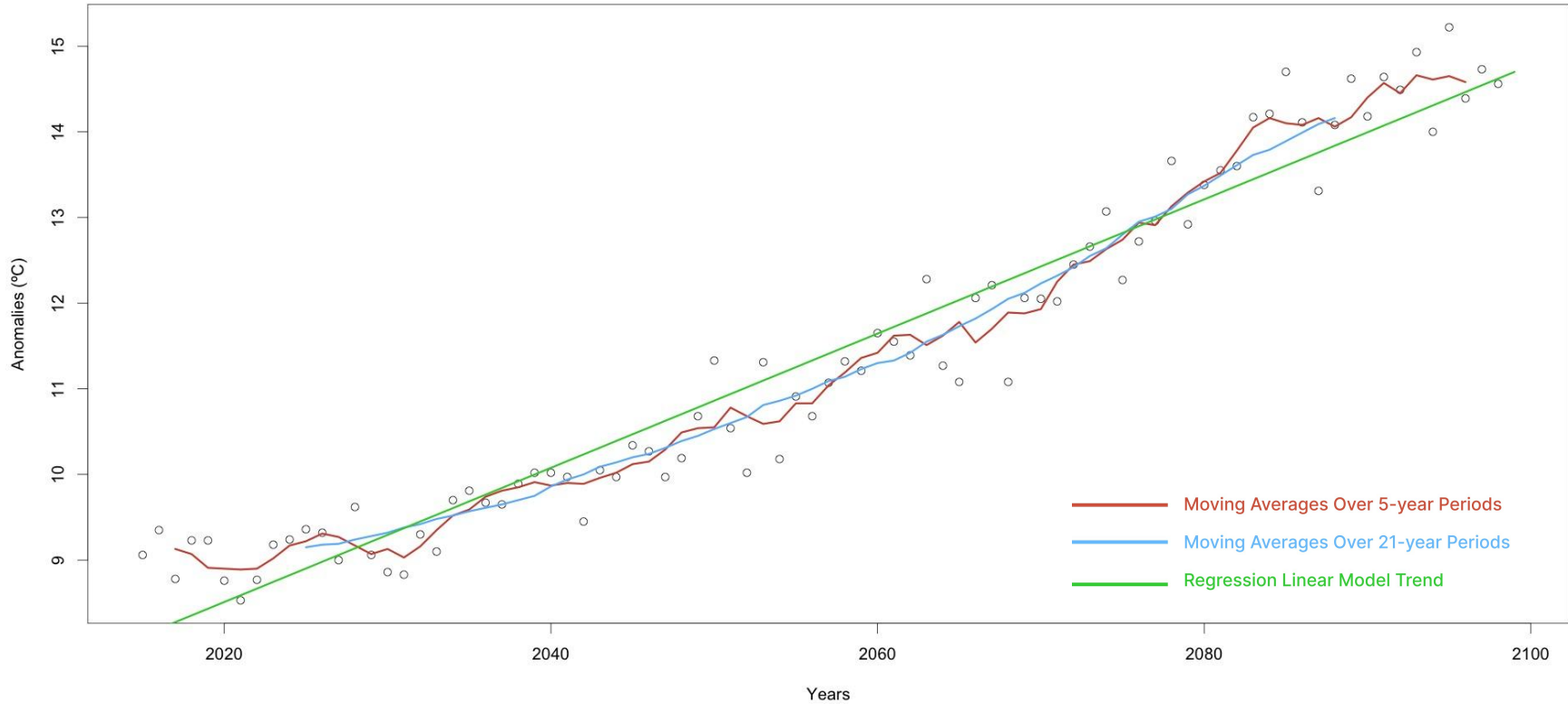
Image source: Google Earth

Yearly Average Anomalies (SSP1-1.9)



- The yearly average temperature anomalies show a **gradual increasing** trend over the study period (2015-2099).
- The anomalies fluctuate around 9°C in the initial years, reaching a maximum of around 10.6°C towards the end of the century.
- There are some notable peaks and troughs, but the overall trend is a **steady rise** in temperature anomalies.

Yearly Average Anomalies (SSP5-8.5)



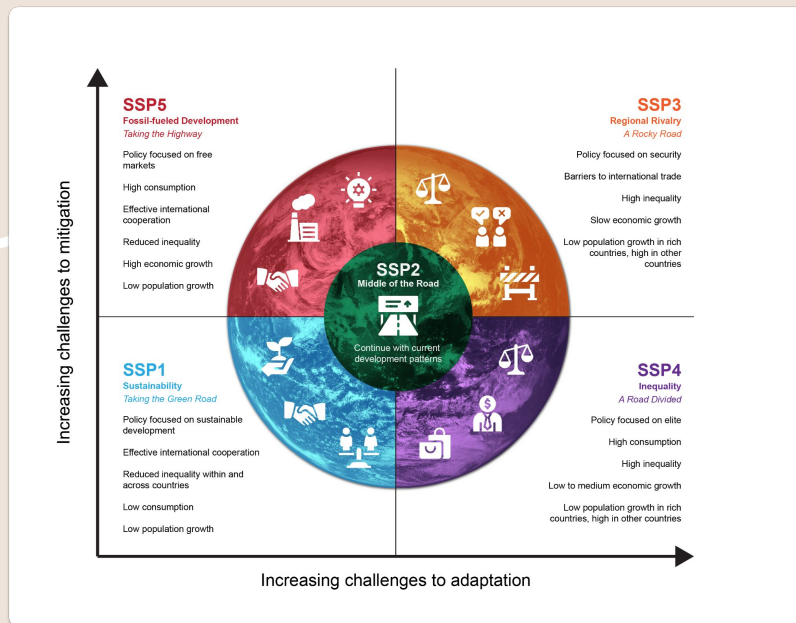
- The yearly average temp. anomalies exhibit a more **pronounced** and **steeper increasing trend** compared to the SSP1-1.9 scenario.
- The anomalies start around 9°C in 2015 but **rapidly** increase, reaching values above 15°C by the end of the century.
- The anomalies rise appears to accelerate **after the mid-century mark** , indicating a **more severe warming trend** under this scenario.

Interpretation of Results

The analysis of the yearly average temperature anomalies under the two scenarios (SSP1-1.9 and SSP5-8.5) reveals **significant differences** in the projected temperature trends over the study region.

Under the SSP1-1.9 scenario, which represents a more sustainable and low-emissions future, the temperature anomalies show a gradual increase. This scenario suggests a moderate warming trend, potentially manageable with appropriate adaptation measures.

In contrast, the SSP5-8.5 scenario, which represents a high-emissions future with continued reliance on fossil fuels, exhibits a much more severe and accelerated warming trend.



Thank you!

Any questions and discussions are welcome.

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